



Containerization vs Virtual Machines

What Are Virtual Machines (VMs)?

- A Virtual Machine is a complete simulated computer running on top of a physical machine.
- It includes its **own OS (Guest OS)**, binaries, libraries, and applications.
- Uses a **hypervisor** (like VMware, VirtualBox, Hyper-V, KVM) to share hardware resources between multiple VMs.
- VMs are **heavyweight** because each VM contains a full OS image.



What Is Containerization?

- Containers package an application along with only the necessary dependencies.
 - They **share the Host OS kernel**, meaning no separate guest OS is required.
 - Tools like **Docker**, **containerd**, and **Podman** create and run containers.
 - Containers are **lightweight**, portable, and start extremely fast.
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Architecture Difference

Virtual Machines

Hardware → Hypervisor → Guest OS → Libraries → Application

Containers

Hardware → Host OS → Container Engine (Docker) → App + Dependencies

Containers skip the “full OS” layer — this is the biggest reason for speed and efficiency.

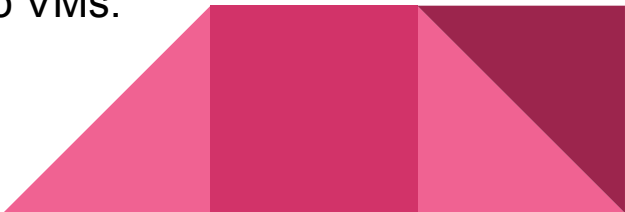


Performance & Resource Usage

VMs

- Slower boot time (30 sec to minutes).
- High resource usage because each VM runs a complete OS.
- Good for **running different OS types** on the same hardware.

Containers

- Start in seconds or milliseconds.
 - Use fewer CPU/RAM resources.
 - More containers can run on the same machine compared to VMs.
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Use Cases

Virtual Machines

- When you need **full OS isolation**.
- Running **different OSes** (Linux, Windows, etc.).
- Strong security boundaries.
- Ideal for **legacy applications**.

Containers

- Microservices architecture.
 - CI/CD pipelines and DevOps automation.
 - Quickly scaling applications.
 - Cloud-native applications (Kubernetes, ECS, EKS, GKE).
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